7.3.2 Structure of Outer Membrane

Eubacteria and Archaeobacteria (Gram positive and Gram negative) differ with respect to their cell walls. Gram negative cell walls are more complex. An outer membrane surrounds a thin underlying layer of peptidoglycan (Table 7.4). Outer membrane is bilayered, consisting mainly of phospholipids, proteins and lipopolysaccharide (LPS).

LPS is composed of three parts which are covalently linked to each other. They are

1. **Lipid A** which is firmly embedded in the membrane,
2. **Core polysaccharide** that is located at the membrane surface and
3. **Polysaccharide O antigens** that extend like whiskers from the membrane surface into the surrounding medium

Special protein channels called porins span the membrane. The points of contact between outermembrane and cytoplasmic membrane are known as adhesions. Outer membrane is anchored to peptidoglycan layer by means of **Braun's** lipoprotein. Periplasmic space between the cell membrane and the outer membrane.

Functions of outer membrane

- It serves as an impermeable barrier to prevent the escape of important enzymes (such as those involved in cell wall growth) from the periplasmic space.
- It serves as a barrier to various external chemicals and enzymes that could damage the cell. For example, the walls of many Gram positive bacteria can be easily destroyed by treatment with an enzyme called lysozyme, which selectively

Table 7.4: Difference between Gram positive and Gram negative bacteria

	Gram positive bacteria	**Gram negative bacteria**
Gram reaction	The bacteria that retain the colour of the primary stain (crystal violet) are Gram positive	The bacteria that cannot retain the primary stain but takes on the colour of the counterstain safranin are called Gram negative
Cell wall	The cell wall is thick (20-30nm thick)	The cell wall is thin (8-12nm thick)
Peptidoglycan layer	Thick (multilayered)	Thin (single layered)
LPS content Lipopolysaccharide	None	High
Periplasmic space	Absent	Present
Outer membrane	Absent	Present
Lipid and lipoprotein content	Low (acid fast bacteria have lipids linked to peptidoglycan)	High due to the presence of outer membrane
Teichoic acids	Present in many	Absent
Example:	*Streptococcus, Staphylococcus, Corynebacterium, Bacillus, Clostridium*	*Escherichia coli, Pseudomonas, Haemophilus, Salmonella, Shigella.*

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